

Learning to Draft: Evaluating Sequential vs. Permutation-Invariant Models for Predicting Professional League of Legends Drafts Under Rule Changes

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Abstract—The draft phase in professional League of Legends involves alternating pick-ban decisions that reflect both game rules and team strategies. This study evaluates two representations for next-action prediction: a sequential formulation modeled with a Long Short-Term Memory (LSTM) network and a permutation-invariant state formulation modeled with a feedforward neural network. Using 54,210 professional games from 2020–2025, we assess model performance under standard drafting rules and the recently introduced Fearless Draft format. When trained on both formats, the LSTM achieves slightly higher accuracy. When trained on regular drafts and tested on Fearless Draft, both models experience substantial degradation, with the LSTM failing to generalize across games in a series, while the set-based model exhibits partial robustness.

Index Terms—Esports analytics, draft prediction, permutation-invariant models, sequence modeling, domain adaptation

I. INTRODUCTION

League of Legends is a globally popular multiplayer online battle arena (MOBA) game played by two teams of five players. Each player on a team controls a unique “champion” that defines their available set of moves in the game. The game supports one of the largest esports ecosystems in the world, with various professional leagues internationally. These leagues run structured regular seasons, playoffs, and international events like the Mid-Season Invitational and World Championship. At this competitive level, even small strategic decisions, such as which champions to select or deny from the opponent, carry a measurable impact on the probability of winning.

One of the substantial processes in the play is the drafting process, a phase for teams alternately ban and pick champions before the game. A ban makes a champion unavailable for both teams during that game, while a pick assigns a champion to its designated team’s composition. The draft progresses in a fixed turn order and is highly strategic, meaning the teams must consider champion familiarity, effectiveness with and against the champions available, and the general strength of the champion on the specific “patch” (game version). Since each

champion can only be selected once per game, teams must anticipate and react to the opponent’s choices in real time. Therefore, modeling this process is a structured prediction problem in which the “next action” (pick or ban) depends on the evolving state of the draft.

There are two natural ways to represent this drafting process for machine learning: as a sequence of actions, or as a state describing all relevant information for the game at a given step. A sequential representation encodes the context for a draft choice as the sequence of picks or bans that came before it. Instead, a state-based representation encodes the context for a draft choice as a permutation set of champions picked or banned, where the champions within those sets are permutation invariant. Intuitively, the state captures nearly all the legally relevant information—if two drafts reach the same configuration of available and unavailable champions, then the optimal next action should, in theory, depend only on that state, not the exact order by which it was reached. This is analogous to chess: while many move sequences can lead to the same board position, the tactical strength of that position does not depend on the path taken.

However, the sequence may still contain strategic information that the state alone cannot express, that the order of the bans can reveal which champions a team fears the most, and the timing of the picks can indicate priorities, comfort preferences, and psychological pressure responses. As a result, while a permutation-invariant state representation is theoretically sufficient to capture the rules of the game, a sequential representation may uncover richer behavioral patterns that teams demonstrate during real drafts.

Competitive League of Legends is frequently played in multi-game series, such as best-of-three or best-of-five formats. Traditionally, teams may reuse champions between games; the only constraints apply within each game’s draft. However, in March 2025, many leagues introduced Fearless Draft, a rule in which champions used by a team in an earlier game of a series become unavailable to the same team in

later games. This rule fundamentally changes the dynamics of the draft by turning earlier champion choices into series-level implicit bans [1]. Fearless Draft therefore creates a meaningful distribution shift relative to traditional drafting: not only are more champions removed from the pool as a series progresses, but teams must also plan across games rather than treating each game independently.

Our work aims to model drafting behavior under standard rules and evaluate different representations. We hope to evaluate if draft order affects decision making for professional League of Legends teams. Additionally, we hope to evaluate how the structural domain shift under Fearless Draft affect robustness the robustness of models developed under the two models.

II. RELATED WORKS

As the popularity of MOBA games grows, the number of studies related to this field is increasing. With the numerous data generated, affiliated parties, including professional teams, companies, and enthusiasts keep expressing their interest in the predictions on the games' outcome, and any predictions related to the outcome itself, thus research and studies related to this field have seen substantial growth in their significance.

A. Machine Learning in Esports and MOBA Games

There are many studies on machine learning predictions, and the perspectives vary in their exploration of the relations. Costa et al. investigated whether information from the picks and bans phase in professional League of Legends matches can be used to reliably predict match outcomes [2]. They applied several supervised learning algorithms to evaluate how the features are related to the games' outcome, and the ban-pick feature is one of them. However, they did not have any further analysis on the prediction of the states in the ban-pick phases. Arik et al. investigated the use of machine learning models to predict the winner of League of Legends matches, but the research is more related to the player performance metrics than the draft composition [3]. Mortelier & Rioult introduced a unique perspective on the analysis of MOBA games; on the other hand, they reduced the complexity of analyzing MOBA games by focusing on simple geometric clues derived from player trajectories, rather than players' statistics or skill trees [4].

B. Drafting and Recommendation Systems

In the field this project focuses on, there are also many research approaches in drafting composition and recommendation systems. Chen et al. used a method that combines Monte-Carlo Tree Search and Neural Networks to find optimal ban-pick decisions under combinatorial explosion conditions [5]. However, they fail to account for critical pre-draft dynamics, such as banning heroes, and often struggle to adapt to dynamic changes in game meta or diverse player styles. Gourdeau & Archambault introduced an automated drafting assistant for professional MOBA games that can recommend optimal hero selections during the ban-pick phase; they implemented discriminative Neural Networks to demonstrate that automated

systems can approximate professional drafting strategies, even across multiple patches and maps, meaning the direction of this project has the possibility to lead to meaningful results [6]. A closer approach is found in the study by Inzitari et al., where they also introduce a mechanism to predict the champion of the next ban-pick phase, using a Recurrent Neural Network (RNN), Long Short Term Memory Network (LSTM), Bi-directional Long Short Term Memory Network (BiLSTM), and Convolutional LSTM [7]. However, their approach uses the sequence of previous phases, meaning each hidden state needs to be maintained across time; thus, such an approach would be more computationally expensive. Hong et al. predict the champions a team should select in the League of Legends in sequential order that aligns with the pick-ban process in the game [8]. However, this approach must train separate models for each team, meaning it does not fully account for interactions between the two sides' choices. It also lacks evaluating the future consequences of that choice dynamically. In short, prior systems treat drafting as sequential or search problems; they are sensitive to order and patch drift. This project reframes drafting as permutation-invariant state prediction, collapsing equivalent sequences.

C. Permutation-Invariant Representation Learning

The research related to permutation-invariant representation learning was also studied prior to the start of the project. Deep Sets, proposed by Zaheer et al., is one of the latest technologies based on this topic; it is a neural architecture for learning on unordered sets where the output must be permutation-invariant [9]. They prove a general form for such functions and show that their simple "embed $\rightarrow$ sum $\rightarrow$ transform" design works well across tasks like point-cloud classification and set-based prediction. Wagstaff et al. further analyze Deep Sets and prove that Deep Sets is only guaranteed to represent and approximate all continuous set functions if the latent dimension is at least the set size, and interpret Deep Sets as a special, most compact case of the broader Janossy Pooling Framework, with implications for attention-based and other set models [10]. Kalra et al. also propose a model based on Deep Sets; the model is a Memory-based Exchangeable Model (MEM), a permutation-invariant architecture that uses memory blocks and self-attention to learn dependencies within sets and unordered sequences; it further enhances the robustness of the permutation-invariant architecture on the tasks where inter-instance relationships within the set are crucial [11]. Kortvelesy et al. introduce PISA, a permutation-invariant set autoencoder that maps variable-sized sets into fixed-size embeddings via a key-value style encoder-decoder [12]. PISA achieves low reconstruction error and preserves similarity in latent space, and is used as a core component in a fusion Graph Neural Network (GNN) for multi-agent sensor fusion and communication. In addition, Balan et al. studied permutation-invariant representations from a geometric perspective by constructing Euclidean embeddings of matrices modulo row permutations [13]. They consider symmetric-polynomial features and a novel sorting-based map, proving

conditions under which the embedding is bi-Lipschitz and showing how nearly any permutation-invariant classifier can be implemented with negligible loss. Their experiments on graph learning benchmarks (e.g., QM9, PROTEINS_FULL) further support the practical value of carefully designed permutation-invariant embeddings. Together, these architectures suggest a spectrum of design choices for set-structured data: from simple sum-pooling (Deep Sets) to memory-augmented and autoencoding models (MEM, PISA). Our work situates League of Legends drafts within this set-learning landscape, where a draft can be represented either as an ordered sequence of picks and bans or as a set/multi-set of champion selections.

D. Domain Adaptation and Patch Dynamics

Domain adaptation and patches would be one of the focuses of this project, as challenges in esports, including frequent balance patches, evolving metas, and regional differences, would cause changes in the data, thus affecting the accuracy of the models that cannot generalize this aspect. There are existing solutions that can overcome such challenges. Huang designs a scalable, adaptive system that recommends optimal ban-pick strategies in competitive MOBA games, using Quantum-Inspired Multi-Agent Reinforcement Learning, meta-learning, GNNs, and multi-level game tree optimization [14]. On the other hand, Liu et al. developed a representation learning method that is permutation-invariant and can be applied to feature selection tasks, especially in settings where data is distributed across clients and privacy must be preserved, to reduce the dependencies on feature ordering and the bias led by correlations [15]. This project would derive the ideology from these solutions to keep the noise of the predictions at the lowest level possible.

E. Synthesis and Gap

The related works reveal both progress and limitations: existing outcome prediction studies show that reliable predictions depend mostly on player-champion performance histories, but not on the prediction of draft states themselves; drafting assistants and sequence-based models approximate professional strategies, yet they inflate the state space by treating equivalent sequences as distinct, thus negatively affecting computational efficiency; permutation-invariant architectures prove that sets can be modeled in a principled way, but they have not been applied to the unique structure of esports drafts; domain adaptation methods demonstrate how models can remain robust under distributional shifts, but they have not been tailored to the frequent balance patches and regional metas that characterize competitive League of Legends. In addition, the newly introduced Fearless Draft mechanism further makes this topic unique among the research topics for machine learning in MOBA games.

Therefore, the gap lies in the absence of a framework that treats draft states as permutation-invariant sets, embeds categorical context such as team, patch, and phase, and adapts across evolving metas. This project addresses the gap by interpreting drafting as a permutation-invariant state prediction problem,

merging equivalent sequences into a single representation, and integrating domain adaptation strategies to ensure robustness across patches and regions.

III. METHODS

A. Problem Formulation

In a League of Legends "draft", opposing teams select ("pick") or eliminate ("ban") champions across twenty phases for each "game" (Fig. 1). A professional "match" may consist of up to five games, each with its own draft. Matches using the "Fearless Draft" format imposes an additional constraint: any champions previously picked in a previous game will be automatically banned in the subsequent games of the series. In standard formats, drafts are independent across all games in the match. To differentiate between these structures, we treat each game in a match as a separate, standalone match in standard formats.

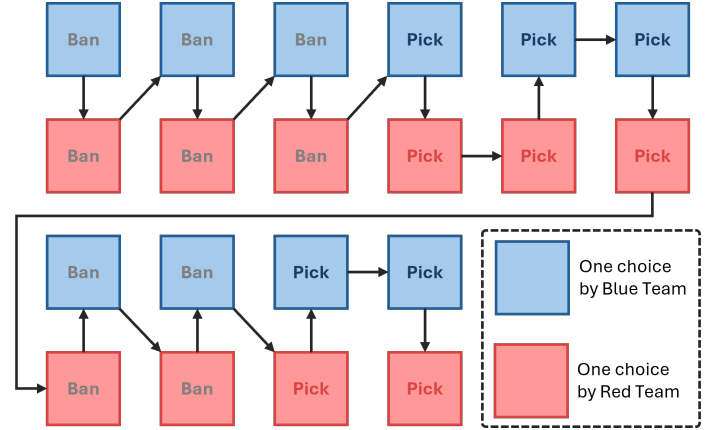


Fig. 1. The Ban-pick Process.

To investigate draft prediction, we consider two representations for the context of draft decision.

R1: A draft is modeled as a sequence of up to 100 draft decisions, each encoded as a one-hot vector. The goal is to estimate the next decision given the observed sequence so far, along with contextual metadata such as the teams, game version ("patch"), the acting team in team in the next phase, and the action type (pick or ban). Formally, for a sequence

$$X = (x_1, x_2, \dots, x_t), \quad x_i \in 0, 1^C, \quad (1)$$

we seek a function

$$f_{\theta}(x_{1:t}, \mathbf{m} * t) \rightarrow x * t + 1, \quad (2)$$

where $\mathbf{m}_t$ is the metadata at phase t .

R2: A draft is modeled as three multi-hot vectors that represent champions in the blue pick space, red pick space, and ban space. The goal is to estimate the next decision with the same contextual metadata, but instead with the state immediately preceding the target. Formally, we seek a function

$$g_{\theta}(\mathbf{b}_t, \mathbf{r}_t, \mathbf{s}_t, \mathbf{m} * t) \rightarrow x * t + 1, \quad (3)$$

where $\mathbf{b}_t, \mathbf{r}_t, \mathbf{s}_t \in 0, 1^C$ are the pick/ban sets at phase t .

B. Research Objectives

To understand how these representations behave under different scenarios, we have the following hypothesis and corresponding objectives

Hypothesis: We hypothesize that a model trained on the set-based representation will be on-par with or marginally worse than the best model trained with the sequential representation. However, we hypothesize that the set-based representation will be far more effective at generalizing to the Fearless Draft format without prior training on it.

O1: Compare the performance of R1 (sequential) and R2 (set-based) using the two models on the entire dataset.

O2: Assess the ability of each representation to generalize from training data without Fearless Draft to test data that with Fearless Draft.

C. Research Methodology

1) *Learning Pipeline:* Our dataset contains 54,210 professional League of Legends games played between 2020 and 2025. Each game was annotated with the patch, the teams on either side, and the sequence of champions picked and banned during the draft. Because not all games are multi-game series, Fearless Draft is a relatively new format, and we treat each game in a multi-game series as a standalone match under standard drafting rules, the majority of matches in the dataset consist of a single game.

TABLE I
GAMES PER MATCH

Games	1.0	2.0	3.0	4.0	5.0
Matches	45828	1443	1181	272	513

Across our dataset, there are 134 patches, 1,301 professional teams, and 171 champions. Champion usage in draft is highly imbalanced as the majority of champions appear in less than 10% of matches.

TABLE II
CHAMPION MATCH PRESENCE FREQUENCY

Frequency	0–10%	10–20%	20–30%	30–40%	40–50%	50–60%
Matches	94	35	30	13	1	1

We preprocessed our data by filtering out corrupted drafts in which champions were incorrectly recorded as being picked or banned more than once per game. Then, we mapped all teams, patches, and champions to IDs for the model. For the sequential representation, we constructed a sequence of champion decisions in draft, with each champion encoded as a one-hot vector over the champion space. For our set-based representation, we converted the sequence into three multi-hot vectors for the picks of the blue team, picks of the red team, and the banned picks from both teams. For both representations, each sample is augmented with data on the acting team (red or blue), and action type (pick or ban) for the target sample using two binary flags. To train the

regular models, we split matches 8:1:1 to create train, test, and validation sets respectively for both models. To train on non-fearless data, we split the 45828 matches from matches with only one game for our train set, and allocated 4191 matches with more than one game to each of the validation and test set.

2) Models Considered:

Feedforward Neural Network - A feedforward neural network (FNN) is a supervised learning model from the input layer, through one or more fully connected hidden layers, to an output layer. Each layer applies a linear transformation followed by an activation function, which allows the network to learn relationships between the input features and targets. This is well suited for training on our set-based representations.

For categorical prediction tasks like ours, the output layer usually has one neuron per class, and a softmax activation is applied to convert the network's output scores into a normal distribution over the categories. During training, the parameters of the network are optimized using gradient descent to minimize a loss function such as cross-entropy.

Long Short-Term Memory (LSTM): A Long Short-Term Memory network is a recurrent neural network architecture designed to model sequences. LSTMs use gating mechanisms that control the flow of information, which allows the model to retain context over long sequences. At each time step t , the LSTM maintains a hidden state and a cell state that summarize the information in the sequence processed so far. This is well suited for training on our sequence-based representations.

For categorical prediction tasks like ours, the final hidden state is passed through a fully connected layer followed by a softmax activation to convert the network's output scores into a normal distribution over the categories. The training optimization process is the same as an FNN.

3) *Data Gathering:* The original dataset was compiled from five files in the Oracle's Elixir esports match data dataset for years 2020 - 2025 [16].

4) *Data Analysis:* To evaluate the performance of the models, we use accuracy as our metric. For a dataset with N samples, let the model output for sample i be the vector

$$\hat{\mathbf{y}} * i = [\hat{y} * i1, \hat{y} * i2, \dots, \hat{y} * iK], \quad (4)$$

where K is the number of classes. The predicted class label is obtained by selecting the index corresponding to the maximum value in this vector:

$$\hat{c} * i = \arg \max_k \hat{y}_{ik}. \quad (5)$$

Let c_i denote the true class label for sample (i). Accuracy is then computed as the proportion of samples for which the predicted class matches the true class:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(\hat{c}_i = c_i), \quad (6)$$

where $\mathbf{1}(\cdot)$ is an indicator function that equals 1 when the prediction is correct and 0 otherwise.

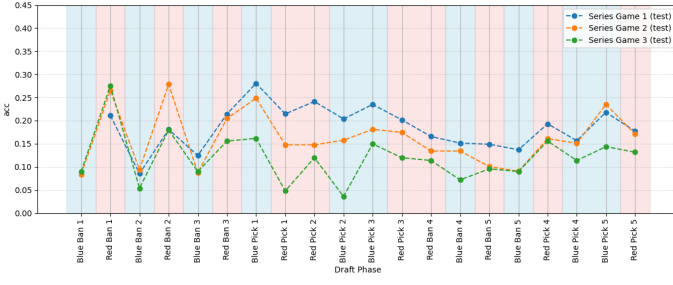


Fig. 2. LSTM Performance over Phases

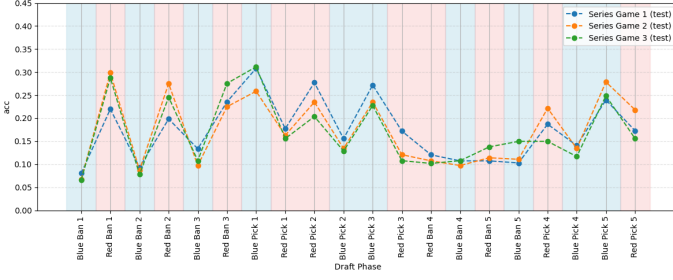


Fig. 3. Neural Network Performance over Phases

5) Limitations and Threats to Validity:

We acknowledge that this dataset and our modeling assumptions are narrowly grounded in the structure of League of Legends drafts. Our set-based representation encodes domain knowledge that is unique to the structure of this game with the way the champions in the sequence are distributed across the three sets. The encoding of the domain knowledge in the set-based representation gives provides it with information that the sequence-based representation does not have access to. However, we believe this to be an intrinsic advantage of the set-based representation.

Furthermore, this means our findings are limited to the domain of League of Legends drafts, and do not generalize to other domains where sequences may be encoded into states with different mapping rules. Thus, the conclusions drawn about the relative effectiveness of the sequential and set-based representations should be interpreted as specific to this domain and not as evidence of general utility in broader sequential prediction tasks.

6) Rationale:

Architectures:

For the set-based representation, we used a multilayer perceptron, with learned embeddings for teams and game versions, followed by two fully connected layers. Team identifiers are embedded with two $n_{teams} \times 16$ dimensional embeddings layers and the patch identifier is mapped using a $n_{patches} \times 12$ dimensional embedding layer. We concatenate these embeddings with three multi-hot champion vectors representing the blue pick space, red pick space, and ban space, together with two binary flags to indicate the acting team and action type in the target phase. The resulting input is then projected to a 512-dimensional hidden layer using ReLU activation, followed by

dropout with a probability of 0.3. A final classifier maps the hidden representation to an output vector corresponding to the champion logits.

For the sequence-based representation, we used an LSTM, combined with metadata embeddings. Team identifiers and patch identifiers are embedded in the same fashion as for the multilayer perceptron, which then are concatenated and projected to a 64-dimensional context vector. Input sequences consist of one-hot champion vectors processed through an LSTM with hidden size 256 using ReLU activation. At each timestep, the LSTM output is concatenated with the context vector and two binary flags to indicate the acting team and action type in the target phase. Finally, dropout layer with a probability of 0.3 is applied. Then, a linear classifier maps the representation to an output vector corresponding to the champion logits.

Parameters:

We optimized our embedding sizes and hidden layer sizes using the Tree-structured Parzen Estimator implemented in the Optuna Python library on our validation set. We train both models with the Adam optimizer use an initial learning rate of $3E-4$ for the Feedforward neural network and $3E-3$ for the LSTM. We track the best model and apply early stopping to our model training when the validation loss increases for 3 epochs in a row.

Models:

We chose a feedforward neural network and LSTM-based model because both models naturally accommodate the structure of our task. Our model needs to take data on teams and patches, of which there are many. Neural networks provide a native mechanism for embedding these categorical signals into feature spaces. This makes neural networks a good fit for our task.

For the sequence-based representation, we chose an LSTM because it is specifically designed to leverage sequential dependencies across the inputs.

D. Results

[/]Experiment 1: Model Performance on Baseline Dataset

Model	FNN	LSTM
Accuracy	0.1759	0.1818

We first evaluated both models on the regular train/validation/test split with both games with both Fearless Draft and standard rulesets. As hypothesized, the LSTM trained on the sequential representation achieves slightly higher overall accuracy than the feedforward model. This is expected, as the sequential representation allows the model to exploit sequential dependencies and ordering cues that the set-based representation cannot directly encode.

Fig 2. and Fig 3 illustrate accuracy per phase across our models. We observe substantial variation in performance between draft phases: some phases, especially the early Red Team pick and bans, are consistently easier for both models to predict. This pattern is stable across both models and games

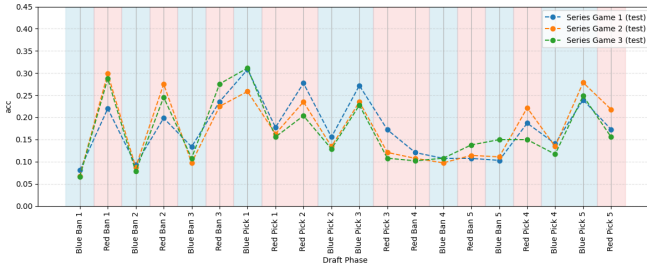


Fig. 4. FNN Performance - Trained on Regular Tested on Fearless Draft

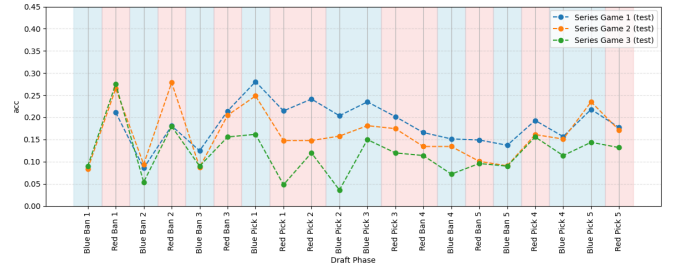


Fig. 5. LSTM Performance - Trained on Regular Tested on Fearless Draft

in a series, suggesting that certain stages of draft inherently exhibit higher decision regularity.

Notably, we also observe the performance of the LSTM model declining later games in a series in a fashion the feedforward model does not. It is possible that due to the lower sample size of series with multiple games, that the LSTM is unable to learn stable temporal patterns for these cases. Though the LSTM performs better overall, it appears to degrade with longer context windows.

[/]Experiment 2: Model Performance on Rule-Split Dataset

Model	FNN	LSTM
Accuracy	0.1154	0.0701

We evaluate model generalization to the Fearless Draft format, where champions selected in earlier games become unavailable in later games of the match.

As seen in Fig 4. and Fig 5., both models exhibit significant degradation in accuracy, even in the first game of a match. There are likely two contributing factors to this. First, professional teams from leagues that predominantly adopted Fearless Draft provide relatively few samples without fearless drafts, producing less effective team embeddings for those teams. Secondly, drafting strategies in Fearless Draft may have diverged from strategies under regular rules, reducing the generalization of patterns learned from conventional datasets.

Despite the degradation, the set-based feedforward model maintains moderate predictive ability across later games in the series. In contrast, the LSTM model trained on sequential data fails to generalize beyond the first game. We can see that for Game 2 and Game 3 of a Fearless Draft series, the model produces effectively random predictions. This highlights a key limitation of sequence-based models: without explicitly defined domain structure, they struggle when the underlying process changes in ways not represented during training.

IV. CONCLUSIONS AND FUTURE WORK

A. Conclusions

Our findings show that drafting order has a measurable but modest effect on strategies employed by professional League of Legends teams. We also find that certain stages of draft exhibit higher regularity than others.

We show that the sequential representation, despite lacking explicit domain knowledge modeled into its structure, achieves

slightly higher performance than the set-based representation when trained on both Fearless and regular drafts.

However, the LSTM model is unable to generalize to Fearless Drafts when only trained on sequences from regular drafts. In contrast, the feedforward neural network trained on set-based representation of regular drafts shows partial robustness though the performance drop remains non-trivial.

B. Future Work

One limitation of our work is in the limited techniques we have tried. Future research could explore using domain adaptation models to improve performance of sequential representations. There is also room to investigate sequence-of-sets representations, which may combine the strengths of both approaches. Such architectures may better leverage the structural rules of drafting while preserving the sequential information.

Beyond drafting, this idea natural extends to other domains involving sequential decisions. Examples could include predicting moves in Go, modeling evolving online shopping carts, or other combinatorial environments where actions accumulate into states representable as permutation-invariant sets.

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